

Mengxuan Helen Qiao

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ABOUT

I am a PhD candidate in the Causal Cognition Lab in the Experimental Psychology Department at University College London. My research investigates judgment and decision-making, Bayesian and causal reasoning, coherence-based reasoning, and AI alignment, with a particular focus on reasoning in legal contexts. Using cognitive science and experimental methods, I investigate how people and large language models evaluate evidence and reach judgments, and I develop Bayesian and computational cognitive models to capture these processes.

EDUCATION

Experimental Psychology, PhD

UNIVERSITY COLLEGE LONDON

Sept 2023 – present

Supervisor: David Lagnado

Crime Science, MSc

UNIVERSITY COLLEGE LONDON

Sept 2021 – Sept 2022

Grade: Distinction

Psychology, BSc

UNIVERSITY COLLEGE LONDON

Sept 2018 – Jul 2021

Grade: 1st Class Honours

RESEARCH

Communicated research in high-impact conference and journal articles across cognitive science, psychology, AI, law, and interdisciplinary venues and presented findings at international conferences (CogSci, ICAIL, SARMAC, JURIX) and universities (UCL, UCLA, University of Edinburgh, University of Groningen, Swansea University).

SKILLS

Technical: R, SPSS, Stata, Python, Minitab, $\text{\LaTeX}$, Qualtrics, Gorilla

Languages: Mandarin Chinese (native), English (full professional proficiency), French (intermediate)

EXPERIENCES

PHD RESEARCHER | UNIVERSITY COLLEGE LONDON

Sept 2023 – present | London, United Kingdom

- Collaborated with an interdisciplinary team of psychologists, cognitive scientists, computer scientists, statisticians, and legal experts, to produce high-impact research, which I communicate at international conferences and in academic journals.
- Developing an integrated cognitive computational model of Bayesian and constraint satisfaction networks to capture the legal reasoning process.
- Developed and tested a novel experimental paradigm to probe cognitive biases in LLMs and compared them to human performance, identifying interaction design levers and enabling systematic testing of AI alignment in legal decision-support and other contexts.
- Designed and supervised a series of experiments examining causal and responsibility attribution to AI systems and associated human actors (developers, users, and other stakeholders), generating insights relevant to AI governance and regulatory policy.
- Contributed to interdisciplinary teams of researchers and legal experts on high-profile criminal appeal cases, and partnered with legal scholars to examine the use and effects of pre-recorded cross-examination under Section 28 of the Youth Justice and Criminal Evidence Act (1999).
- Peer reviewer for journals and conferences, including *ACM CHI*, *CogSci*, *Behavioral Science*, and *Legal and Criminological Psychology*.

VISITING SCHOLAR | UNIVERSITY OF CALIFORNIA, LOS ANGELES, UCLA REASONING LAB

Sept 2025 – Nov 2025 | Los Angeles, United States

- Worked with Professor Keith Holyoak on coherence-based reasoning in legal decision-making and computational modelling of evidential probability judgments.
- Developing a novel computational model formalising how Bayesian probabilities operate within a constraint satisfaction network.

RESEARCH ASSISTANT | QUEEN'S UNIVERSITY BELFAST & SWANSEA UNIVERSITY, HILLARY RODHAM CLINTON SCHOOL OF LAW

Aug 2024 – present | Remote

- Working on the Trust in User-generated Evidence (TRUE) project, which seeks to explore the impact of AI generated misinformation and deepfakes on trust in user-generated evidence in accountability processes for human rights violations.
- Designed a series of experiments examining how non-deceptive AI edits to evidence influence perceived credibility and admissibility.

AFFILIATED RESEARCHER | UNIVERSITY OF GRONINGEN

Aug 2025 – present | Remote

- Working on the Preventing Miscarriages of Justice (PMJ) project, developing an integrated training method of Bayes' rule, scenario theory, and causal Bayesian networks to help judges to reason more rationally about evidence.

SUPERVISOR & PGTA | UNIVERSITY COLLEGE LONDON

Sept 2023 – present | London, UK

- Supervised third-year undergraduate students on their dissertations, and led seminar groups to undergraduate students.